CHAPTER 8

Full calculation

S59, S61, S63, S66

Before you start · Primitives, signs and sums

Without a calculator, in six minutes. Respond briefly. Mobilize areas and derivative of a product.
Then choose the targeted review indicated under the questions.

1. Give a antiderivative of 2 x.

2. Determine the sign of x on $[-1;2]$.

3. Calculate the area of a side rectangle 3 and 5.

4. Derivating xe^x.

5. Write $\sum_{k=1}^{n} k$.

6. Calculate the difference between areas 2 and 5; can it be negative?

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Deriving the answer candidate; R64. **2.** Cut the interval in point 0 and read the sign. **3.** Multiply Lengths and indicate a squared unit. **4.** Write both terms of the derivative of the product; R35. **5.** Count the terms and match first and last. **6.** Perform ordinate subtraction and to distinguish between areas and differences. Questions R serve as support for resuming with the teacher, even if their collective ritual is planned later.

Objectives

Calculate and frame an integral; distinguish area and integral; use integration by study an integral sequence.

Course notes

Air and antiderivative

For f continued and positive on $[a,b]$, $a \leq b$, integral $\int_a^b f(x)\,\mathrm{d}x$ is the area under the curve, above of the axis between $x = a$ and $x = b$, expressed in d ·air units.

Fundamental Theorem

If f is continuous over an interval I and $a \in I$, then $F_a(x) = \int_a^x f(t)\,\mathrm{d}t$ is the antiderivative of f who annule in a. For any antiderivative F,

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Any continuous function over an interval admits antiderivatives. An integral is a number; a antiderivative is a function.

Demonstration in the increasing positive case. For $h > 0$, the rectangles give $hf(x) \leq F_a(x+h) - F_a(x) \leq hf(x+h)$. After division by h the rate of increase tends to $f(x)$. For $h < 0$, growth $f(x+h) \leq f(t) \leq f(x)$ on $[x+h, x]$. By integrating,

$$(-h)f(x+h) \leq F_a(x) - F_a(x+h) \leq (-h)f(x).$$

By dividing by $-h > 0$, we get

$$f(x+h) \leq \frac{F_a(x+h) - F_a(x)}{h} \leq f(x).$$

The two bounds tend to $f(x)$. Out of continuity, $F_a' = f$. Any other antiderivative is $F_a + C$, where the calculation formula at the bounds.

Properties

For continuous functions and bounds within the same range:

$$\int_a^a f = 0, \quad \int_b^a f = -\int_a^b f, \quad \int_a^c f = \int_a^b f + \int_b^c f, \quad \int_a^b (\lambda f + \mu g) = \lambda \int_a^b f + \mu \int_a^b g.$$

If $a \leq b$ and $f \leq g$, then $\int_a^b f \leq \int_a^b g$. In particular, if $m \leq f \leq M$, then $m(b-a) \leq \int_a^b f \leq M(b-a)$. For any sign function, the integral is defined by antiderivatives.
between the curve and the east axis $\int_a^b |f|$, obtained by cutting off the sign changes. Between two curves, it's worth $\int_a^b |f - g|$.

Example. $\int_{-1}^{1} x \, \mathrm{d}x = 0$, but the area between the right and the axis over this interval is 1.

Average value and integration by parts

For $a < b$, the average value is $\mu = \frac{1}{b-a} \int_a^b f(x) \, \mathrm{d}x$. The height rectangle μ and width $b - a$ has the same algebraic area.

Integration by parties

If u, v are of class $\mathcal{C}^1$ on $[a, b]$,

$$\int_a^b u(x)v'(x) \, \mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, \mathrm{d}x.$$

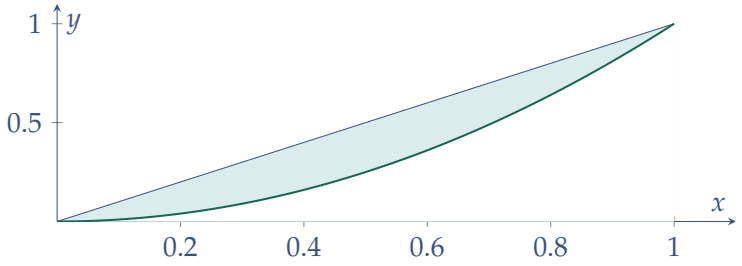
Demonstration. Integrate $(uv)' = u'v + uv'$ then isolate the desired integral. The hook is indispensable.

Example. With $u = x$ and $v' = e^x$, $\int_0^1 xe^x \, \mathrm{d}x = [xe^x]_0^1 - \int_0^1 e^x \, \mathrm{d}x = e - (e-1) = 1$.

Rectangles and integrated sequences

For f continued and increasing on $[a, b]$, $a < b$ and $n \geq 1$ and $h = \frac{b-a}{n}$, amounts $h\sum_{k=0}^{n-1} f(a+kh)$ and $h\sum_{k=1}^{n} f(a+kh)$ The difference between them is $h(f(b) - f(a))$. For $I_n = \int_0^1 x^n f(x) \, \mathrm{d}x$, start with sign and comparisons of integrand. If $0 \leq f \leq M$, then $0 \leq I_n \leq \frac{M}{n+1}$; the gendarmes give $I_n \to 0$. A party-based integration may also provide a relationship between I_n and I_{n+1}.

Convention at zero level. In $I_n = \int_0^1 x^n f(x)\,\mathrm{d}x$, the expression x^0 means constant function 1, y included in bound 0. $I_0 = \int_0^1 f(x)\,\mathrm{d}x$; no undetermined calculation is necessary.



Between $y = x$ and $y = x^2$ on $[0,1]$ value $\frac{1}{6}$.

Linking area, antiderivative and approximation

8.1 Two areas, one integral

First use: S59 and S61.

Example Solved · A Change of Sign

On $[0;3]$, $f(x) = x - 1$ is negative before 1, positive after. $[x^2/2 - x]_0^3 = 3/2$. Geometrical area is $-\int_0^1 f + \int_1^3 f = 1/2 + 2 = 5/2$. Integral compensation between two signed contributions.

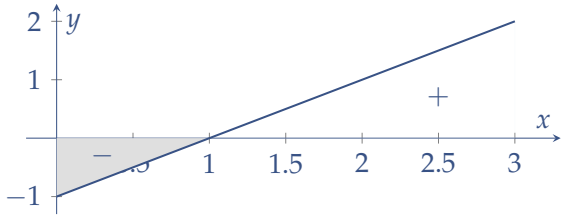


Figure 8.1. For a geometric area, the two regions add up; for integral, their signs Come in.

A1 · Immediate application For $g(x) = 2x - 2$ on $[0;3]$, calculate the integral and the area between the curve What is the relationship with the values of the example?

8.2 Why l ·aire gives a antiderivative

First use: S61.

Ownership · Fundamental Theorem

For f continues over interval I and $a \in I$, $A(x) = \int_a^x f(t)\,\mathrm{d}t$ is a antiderivative f. So, for any antiderivative F, $\int_a^b f = F(b) - F(a)$. The general case is admitted; the positive case increasing is shown below.

Demonstration. Suppose f continues, positive and increasing. For h>0, comparison to two rectangles given $f(x) \le [A(x+h) - A(x)]/h \le f(x+h)$. For h<0, it gives $f(x+h) \le [A(x+h) - A(x)]/h \le f(x)$. By continuity and bounds, the rate of increase tends to f(x); so $A' = f$. Any other antiderivative F differs from A of a constant; like A(a)=0, $A(b) = F(b) - F(a)$.□

Solved Example · Find a Derivative

For $A(x) = \int_0^x 2t\,\mathrm{d}t$, the area is x^2 when x is positive. The formula by antiderivative also gives $A(x) = x^2$ for x negative, with integral orientation. In both cases $A'(x) = 2x$.

A2 · Partial guidance On $[0;+\infty[$, pose $A(x) = \int_0^x e^{-t}\,\mathrm{d}t$. Give A(x), A(x), direction of variation and Interpret the boundary as an area.

8.3. Equivalent height, with its units

First use: S63.

Definition · Average value

For $a < b$, the average of $[a;b]$ is $m = \frac{1}{b-a}\int_a^b f$. If f is positive, the width rectangle b-a and height m has the same area as the area under the curve.

Example Solved · Flow and Volume

A flow rate $d(t) = 2 + t$ L/min for three minutes provides $\int_0^3 (2+t)\,\mathrm{d}t = 10{,}5$ L. Its flow rate Average is 10.5/3 $= 3.5$ L/min. The volume and average flow do not have the same unit.

A3 · Immediate application Calculate the average of x^2 on $[0;1]$. Trace the equivalent rectangle. Explain why this mean is not the average of the two values at the ends.

8.4 Draw and control the frame

First use: S63 and S66.

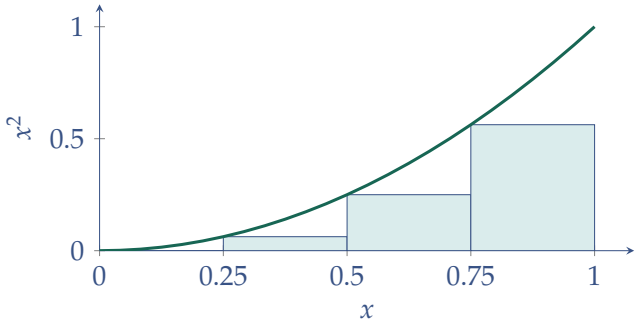


Figure 8.2. Rectangles to the left. Complete the drawing with the rectangles to the right and the right at height 1/3.

Example resolved · Two intervals

For x^2 on [0;1], two rectangles to the left give $L_2 = \frac{1}{2}(0^2 + (1/2)^2) = 1/8$; those on the right give $U_2 = \frac{1}{2}((1/2)^2 + 1^2) = 5/8$. The integral 1/3 is well between these bounds.

A4 · Search For $f(x) = x^2$ on $[0;1]$, cut in four equal intervals. Calculate amounts to the left and to the right; justify their positions in relation to the integral. intervals, their deviation is 1/n.

A first step: In the difference between the two amounts, the internal values are simplified.

8.5 Choosing what's drifting

First use: S66.

Property · Integration by parties

For class u,v $\mathcal{C}^1$ on $[a;b]$, $\int_a^b u'v = [uv]_a^b - \int_a^b uv'$. The formula results from the integration of $(uv)' = u'v + uv'$.

Example resolved · Two successive PPIs

For $I = \int_0^1 x^2 e^x \,\mathrm{d}x$, drifting the polynomial decreases its degree. First $I = [x^2e^x]_0^1 - 2\int_0^1 xe^x \,\mathrm{d}x$. Next $\int_0^1 xe^x \,\mathrm{d}x = [xe^x]_0^1 - \int_0^1 e^x \,\mathrm{d}x = e - (e-1) = 1$. So I=e-2. Write the bounds to each step avoids forgetting a constant.

A5 · Choice of strategy Calculate $\int_1^e \ln x \,\mathrm{d}x$. Justify your choice in IPP and check the result in derived from a antiderivative.

8.6 Frame before looking for a formula

First use: S66.

Solved Example · A Framework Without Explicit Primitive

Poisons $J_n = \int_0^1 x^n/(1+x) \,\mathrm{d}x$. Like 1 $/2 \leq 1/(1+x) \leq 1$, $1/[2(n+1)] \leq J_n \leq 1/(n+1)$. The two bounds tend towards 0, so $J_n \to 0$, without calculating Jn for each n.

A6 · Summary For $n \in \mathbb{N}$, $I_n = \int_0^1 x^n e^{-x} \,\mathrm{d}x$. Show that (I_n) is positive and decreasing. $e^{-1}/(n+1) \leq I_n \leq 1/(n+1)$, then its limit. Does an IPP give a relationship between I_n and I_{n+1} ?

A7 · Refuter Give a non-zero continuous function on $[-1;1]$ of which the integral is null. In addition, with continuity, it would be possible to conclude that an integral function Is zero identically zero?

Wait a minute at the courthouse. Explain on the figure why "integer null" and "area null" do not say Not the same thing.

Review postponed · S79. Calculate an area between trigonometric curves by keeping the search intersections and sign.

Research activity

Frame an integral with rectangles · 35 min.

We're trying to frame $I = \int_0^1 x^2 \,\mathrm{d}x$. For an integer $n \geq 1$, we share $[0;1]$ in n same intervals length. L_n the sum of the areas of rectangles whose heights are the values of x^2 to left ends, and U_n the one obtained at the straight ends.

1. For $n = 4$, represent the two families of rectangles, then calculate L_4 and U_4.

2. Justify that $L_4 \leq I \leq U_4$.

3. Write a Python argument function n which returns L_n and U_n. Algebraically determine $U_n - L_n$.

4. Determine the smallest integer n for which this framework has a width of not more than 0.01.

5. Calculate the exact value of I with the help of a antiderivative and verify that it belongs to the framework obtained for $n = 4$.

Progressive exercises

E1 · Direct application. Calculate $\int_0^2 (3x^2 - 2x + 1)\,\mathrm{d}x$ by determining a antiderivative.

E2 · Application and justification. In an orthonormal coordinate system, calculate the area of the delimited domain by curves $y = x$ and $y = x^2$, between abscisses 0 and 1. Give the result in a unit of ,.

E3 · Method to be organised. Calculate the mean value of the exponential function over the interval $[0;2]$.

E4 · Choice of strategy. Using part integration, calculate $\int_1^e \ln x\,\mathrm{d}x$.

E5 · Synthesis or demonstration. For every $n \in \mathbb{N}$, we're laying $I_n = \int_0^1 \frac{x^n}{1+x}\,\mathrm{d}x$, $x^0 = 1$ in Integrate. Demonstrate that 0 $\leq I_n \leq \frac{1}{n+1}$, then deduct the limit of (I_n).

Work in the manual or annal

Barbazo, p. 260, Exercises 28 and 29. In S66: Exercise 28, clarifying the two functions chosen
At home: exercise 29, to adapt this choice to a product containing
A logarithm.

Duty at home

DM8 · A complete sequence · 60 min.

For every $n \in \mathbb{N}$, we're laying $I_n = \int_0^1 x^n e^x\,\mathrm{d}x$. For $n = 0$, integrande is the function $x \mapsto e^x$.

1. Calculate I_0then I_1 with the help of party integration.

2. Demonstrate that, for everything $n \in \mathbb{N}$, $I_{n+1} = e - (n+1)I_n$.

3. By comparing the integrands, demonstrate that (I_n) is positive and decreasing.

4. Show that $\frac{1}{n+1} \leq I_n \leq \frac{e}{n+1}$ for every $n \in \mathbb{N}$. Deduct the limit of (I_n).

5. In the calculation of I_{n+1} by the relationship of question 2, by what factor an error on I_n Is she If the other errors are neglected, explain the risk of a repeated calculation of the that way.

MCQ A · integral, area and medium

After S63; we always assume $a < b$.

For each question, only one proposal is accurate.
consult the correction.

1. If $F' = f$ on $[a;b]$, then $\int_a^b f(x)\,\mathrm{d}x$ is valid for:

A. $F(a) + F(b)$ B. $F(b) - F(a)$

C. $f(b) - f(a)$ D. 0

2. Integral of a negative function on $[a;b]$ is:

A. Positive B. None

C. Negative or nil D. Always a positive area

3. The average value of f on $[a;b]$ is:

A. $\frac{1}{b-a}\int_a^b f$ B. $\int_a^b f$

C. $f(a) + f(b)$ D. $b - a$

4. The geometric area between the curve and the axis of the abscisses requires:

A. to keep the signs B. cutting when the sign changes

C. a second derivative D. a random drawing

5. A antiderivative of e^x is:

A. xe^x B. e^x

C. $\ln x$ D. $1/e^x$

MCQ B · PPI and management

After S66.

For each question, only one proposal is accurate.
consult the correction.

6. For integration by parties, one must know:

A. Two grade functions $\mathcal{C}^1$ on $[a;b]$ B. a geometric sequence

C. a fixed point D. A binomial distribution

7. $\int_a^b u'v\,\mathrm{d}x$ is equal to:

A. $[uv]_a^b - \int_a^b uv'\,\mathrm{d}x$ B. $\int_a^b uv\,\mathrm{d}x$

C. $[u+v]_a^b$ D. 0

8. If f is increasing and positive on $[0;1]$, the rectangles on the left give :

A. a upper bound B. A minor

C. Always the exact integral D. a negative value

9. A complete sequence $I_n = \int_0^1 x^n f(x)\,\mathrm{d}x$, f continued positive on $[0;1]$, may often be framed using:

A. $0 \leq x^n \leq 1$ B. $x^n = n$

C. $\ln x \geq 0$ D. the derivative of n

10. An integral flow unit (L/min) over a time in min is:

A. $\mathrm{L/min^2}$ B. L

C. $\mathrm{min/L}$ D. Ununited

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. An integral is always positive.

2. Two antiderivatives of the same function over an interval differ from one constant.

3. An approximation by rectangles does not control the direction of the frame.

Review review

I can interpret sign, area and medium; choose a antiderivative or an IPP and frame. **To be monitored:** confounding geometrical and integral algebraic area; forget the bounds in $[uv]_a^b$.